

组合起来

1. Layer ~~can~~ Compact CNN -- init -- (combine the different layers of
self : conv1 = conv2D(, , , conv1,2, pool1,2, relu1,2,
Flatten, Linear)

#RG15

in - channels = 3

out - channels = 8

kernel - size = 3, H - conv kernel = size

stride = 1, out - channels

padding = 1, in - channels

input - shape (N, 3, 32, 32) $\xrightarrow{\text{conv1}}$ (N, 8, 32, 32)

$$\text{Hout} = \frac{CH + OP - K}{S} + 1$$

2- Layer ReLU

for all element from conv1 do $\max(x, 0)$

shape remains

unchanged.

3- Layer Pool

kernel size = 2

stride = 2

Let the space reduce half

for each channel do 2×2 max pooling, stride = 2

so: shape:

$(N, 8, 32, 32) \rightarrow (N, 8, 16, 16)$

4. Layer Conv 2

self.Conv 2 = Conv2D (

in_channels = 8,

out_channels = 16, def by me

kernel_size = 3

stride = 1

padding = 1,)

(N, 8, 16, 16) $\xrightarrow{\text{Conv2}}$ (N, 16, 16, 16)

5. Layer ReLU2

all element do $\max(x, 0)$ one by one.

shape remains unchanged.

6. Layer Pool 2

self.pool2 = MaxPool2D (kernel_size=2,
stride=2)

(N, 16, 16, 16) $\rightarrow$ (N, 16, 8, 8)

7. Layer Flatten

$\uparrow 16 \times 8 \times 8 = 1024$

(N, 16, 8, 8) $\rightarrow$ (N, 1024)

8. Layer classifier / Linear

in_features = $16 \times 8 \times 8$

out_features = 10

$\rightarrow$ why 10?

(N, 1024) $\rightarrow$ ~~(N, 10)~~ because CIFAR-10 has 10 types.

the final output of convnet forward is logits

it isn't probabilities

logits is scores from the model for 10 types
(def in lecture)

e.g. a photo

1.2, -0.5, 0.3, 2.1, -1.0, 0.0, 0.0, -0.2, 1.5, 0.4

softmax classifier output $\rightarrow$ loss